Multilevel Class Token Transformer With Cross TokenMixer for Hyperspectral Images Classification

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***Abstract*— The transformer has become a prominent technique for hyperspectral image (HSI) classification, attributed to its capability to model global dependencies between features. Nevertheless, the predominant transformer-based methods rely on a direct information flow with a fixed number of tokens, causing the sequential transformer encoders to lack crucial interaction. This deficiency results in an inappropriate granularity of discriminative features and the loss of subtle patterns. In response to this limitation, we introduce a novel approach named multilevel class token transformer (MCTT) with cross tokenmixer for HSI classification. Specifically, we explore a convolutional neural network (CNN) stem network that incorporates 3-D, 2-D, and pointwise convolutions to encode local spatial–spectral information. The spectral–spatial features undergo transformation into semantic tokens using a semantic tokenizer. These tokens are then input into the transformer encoder to capture global interactions between different pixels. To create a hierarchical semantic representation, we propose a cross tokenmixer that integrates different levels of class tokens and patch tokens, enabling a multigrained representation. The cross tokenmixers, with their varied number of tokens, facilitate the learning of distinct discriminative spectral–spatial representations and enable a comprehensive understanding of the HSI through a voting mechanism. Extensive experiments and ablation studies are conducted on three public HSI datasets to evaluate the performance of our proposed method. The results demonstrate the effectiveness and superior performance of our approach in HSI classification.**

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***Index Terms*— Convolutional neural networks (CNNs), cross tokenmixer, hyperspectral image (HSI) classification, multilevel class token transformer (MCTT).**

I. Introduction

HYPERSPECTRAL image (HSI) is a type of remote sensing image that combines spectroscopic and spatial features, playing a critical role in the classification and recognition of complicated remote sensing scenes [1], [2]. In comparison to traditional optical images, HSI records a large number of spectral bands, which facilitate elaborate land use/land cover classification. The spectral information in HSI data reveals differences in physical structure and chemical composition, while the spatial information reflects geometric quality features such as shape, size, location, and defects. By combining these pieces of information, the recognition accuracy of ground targets can be significantly improved. Therefore, HSI classification is widely used in various fields such as mineral detection [3], environmental monitoring [4], [5], [6], and agricultural analysis [7].

Machine learning methods have achieved tremendous success in HSI classification. In the early days, techniques such as support vector machines (SVMs) [8], [9], and random forests (RFs) [10] were commonly employed for HSI classification. In addition, some dimensionality reduction methods were utilized to mitigate the computational burden arising from the high-dimensional nature of HSI data, including principal component analysis (PCA) [11], independent component analysis (ICA) [12], and linear discriminant analysis (LDA) [13]. These methods have played crucial roles in HSI classification research. However, with the continuous advancement of deep learning technology, new methods and algorithms have emerged, yielding even better classification results.

Recently, deep learning methods have made significant breakthroughs in the field of HSI classification, harnessing their powerful representation learning capabilities. Chen et al. [14] introduced the first deep learning method for HSI classification, presenting an auto-stacked encoder. In the following years, numerous deep backbone networks have been successfully applied to HSI classification tasks due to their local feature extraction abilities. Different configurations of convolutional neural networks (CNNs), such as 1-D [15], 2-D [16], and 3-D [17], [18], [19], [20], have been widely employed in HSI classification. Hu et al. [15] proposed a 1-D CNN to extract the spectral features of HSI. Yang et al. [21]

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introduced a dual-branch network structure that combined 1-D and 2-D CNNs to extract joint spatial and spectral features. Zhong et al. [17] proposed a variant of the 3-D CNN structure to effectively extract spectral–spatial features. Roy et al. [22] devised a hierarchical structure that combined 3-D and 2-D CNNs to extract both spatial and spectral features, resulting in improved classification accuracy.

To fully leverage the spatial information extraction capabilities of CNNs, some large-scale frameworks based on multilayer CNN architectures have been proposed. Examples include ResNet [23] and DenseNet [24]. For instance, Yang et al. [25] utilized a dual-path DenseNet, which extracted both spectral and spatially-related features. This approach alleviated problems such as gradient vanishing and overfitting, ultimately leading to improved classification accuracy. While CNNs have shown effectiveness in HSI classification tasks, they encounter difficulties in capturing global properties of sequences and extracting remote dependency relationships in the spectral dimension [26], [27]. These limitations can have an impact on the model's ability to accurately analyze and classify HSI data. Meanwhile, recurrent neural networks (RNNs) [28], generative adversarial networks (GANs) [29], graph neural networks (GNNs) [30], [31], and other techniques [32] are also frequently utilized in the context of HSI classification. While these approaches have demonstrated some efficacy, they did not break through the constraints of local dependencies until the advent of transformers.

Recently, transformer [33] and its variants [34], [35], [36], [37], [38], [39] has become a hot topic in HSI classification and has demonstrated its advantages in capturing global dependencies. Hong et al. [40] proposed the SpectralFormer, which extracts spectral information from adjacent spectral bands and then uses the transformer encoder module for classification. Tan et al. [41] integrated a transformer in transformer (TNT) module to construct a deep network model with the extended morphological contour features tailored for HSI classification. Tang et al. [42] proposed a transformer-based backbone network for HSI classification by employing a stack of spectral attention and spatial attention layers. However, the transformer excels at capturing spectral signatures, but it falls short in extracting local fine-grained features and fails to fully utilize spatial information within the HSI.

By leveraging the strengths of both CNNs and transformers, CNN-transformer-based architecture [43], [44], [45] is explored for HSI classification where CNNs are employed for capturing local spatial features and transformers for modeling long-range dependencies and global context. He et al. [46] employed a deep CNN to extract spatial features, followed by a densely connected transformer for capturing spectral relationships within the sequence. Song et al. [47] proposed a dual-branch framework that combines 3-D CNN and spatial–spectral transformer (SSTN) to jointly extract local and global features from HSI. Zhao et al. [48] exploited a convolution transformer fusion splicing network for HSI classification with a residual splicing convolution block to serialize HSI. However, the CNN-transformer-based methods employ low-level CNN features as input for the transformers, resulting in a lack of semantic information. To address the issue, Sun et al. [49] proposed the spectral–spatial feature tokenization transformer (SSFTT) to address the challenge of tokens from the shallow spatial–spectral features lacking semantic information. The SSFTT leverages semantic tokenization to express the low, middle, and deep semantic features of HSI, thereby enhancing the performance of HSI classification. However, the majority of the aforementioned methods predominantly rely on a straightforward information flow through sequential transformer encoders, which lack interactions between hierarchical features when generating high-level semantic information. Furthermore, as all transformer encoders handle a fixed-sized token input, it becomes increasingly difficult for information to traverse across a broader context, consequently resulting in constrained contextual comprehension.

In order to fully utilize the context information extracted by the transformer encoder in HSI, we propose a multilevel class token transformer (MCTT) by establishing the interactions between patch tokens and class tokens across different levels. A CNN stem network is utilized, incorporating 3-D, 2-D, and pointwise convolutions, to encode the local spectral–spatial information. A semantic tokenizer is used to convert the spectral–spatial features into semantic tokens suitable for the transformer. These semantic tokens are then fed into the transformer encoder to capture global interactions between different pixels. Subsequently, a cross tokenmixer is introduced to fuse the input and output of the transformer encoder, creating a hierarchical semantic representation. Finally, multilevel class tokens are integrated to enable a voting mechanism that facilitates a comprehensive understanding of the HSI.

The main contributions of this article are as follows.

1) A MCTT is proposed for HSI classification by leveraging multiple class tokens to enhance the dependencies between class tokens and multilevel semantic patch tokens. Each class token learns distinct discriminative spectral–spatial representations at different levels, facilitating a comprehensive understanding of HSI.
2) A cross tokenmixer is proposed to establish the interactions between the input and output of the transformer encoder. The mixed tokens consider the context and dependencies across the transformer encoder, aiding in the transmission of spectral–spatial features to a hierarchical semantic representation.
3) The spectral–spatial features are managed through the use of incremental tokens, enabling the capture of coarse-to-fine representation of HSI. Extensive experiments and ablation studies were conducted on three publicly available HSI datasets, effectively demonstrating the effectiveness and superior performance of the proposed method.

II. Methodology

An HSI denoted as $Z \in \mathbb{R}^{H\times W\times D}$, is characterized by its height H, width W, and the number of spectral bands D. The HSI contains a total of $H \times W$ pixels, represented as $Z = \{z_1, z_2, \ldots, z_{H\times W}\}$. Each pixel is associated with a ground truth label denoted as $Y = \{y_1, y_2, \ldots, y_C\}$, where C represents the number of ground truth classes. To reduce the dimensionality of the HSI and extract its essential spectral

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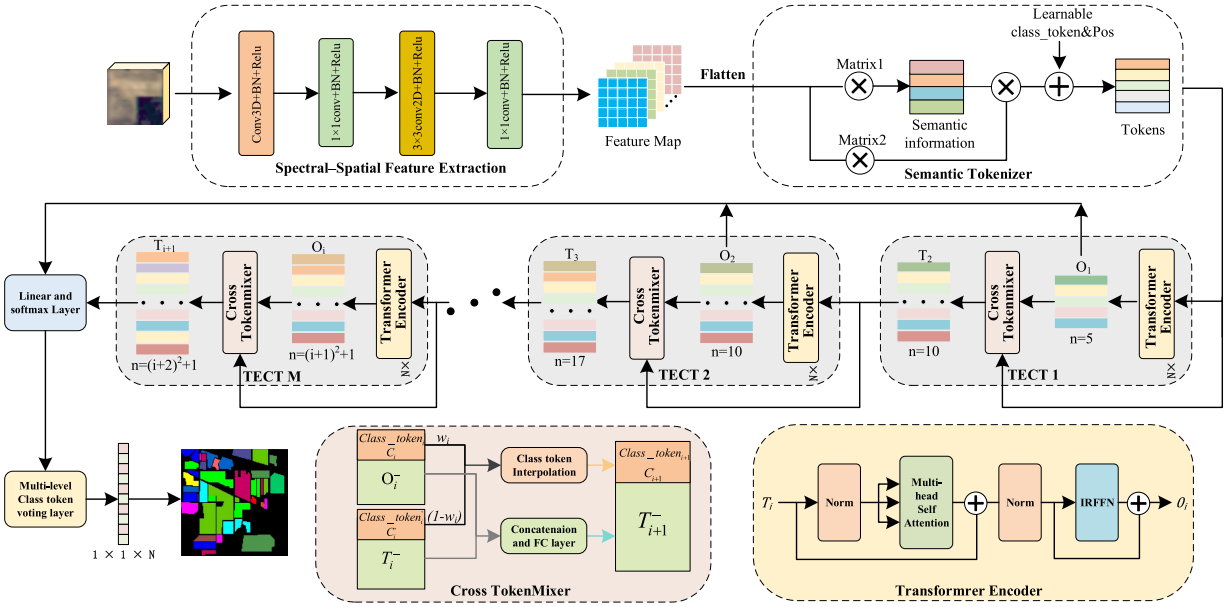


Fig. 1. Multiple convolutional layers are utilized to encode local spectral–spatial features. These features are then transformed into semantic tokens by a semantic tokenizer. The transformer encoders are employed to capture global interactions among the tokens. The cross tokenmixer is proposed to enhance the semantic fusion between different stages of tokens. Furthermore, the incorporation of multilevel class tokens enables a voting mechanism for comprehensive analysis of the HSI data.

information, PCA is performed. This process yields a reduced representation of the HSI denoted as $Z_{\text{pca}} \in \mathbb{R}^{H\times W\times d}$, where d represents the reduced number of spectral bands. In order to leverage both the spatial and spectral information available in the HSI, neighboring 3-D HSI cubes centered at each pixel are utilized for classification. These HSI cubes, denoted as $X = \{x_1, x_2, \ldots, x_{H\times W}\} \in \mathbb{R}^{a\times a\times d}$, have a width of a. The objective of HSI classification is to assign appropriate land-cover labels to each cube x_i based on its spatial and spectral characteristics.

The proposed MCTT for HSI classification is depicted in Fig. 1. The framework consists of several key steps. First, local spectral–spatial features of the HSI are extracted using a CNN block. Second, a semantic tokenizer is applied to the extracted spectral–spatial features to obtain the initial semantic tokens for the transformer. Third, the global features are captured through the interactions of the transformer encoders. Fourthly, the multilevel class tokens and features are fused using the proposed cross tokenmixer, which results in a comprehensive representation of the HSI. Finally, the class labels are determined using a voting mechanism.

A. Spectral–Spatial Feature Extraction

At the initial stage of the network, the training set samples are fed into a series of convolutional layers to extract their spectral–spatial information. The first convolutional layer applied to the samples is a 3-D convolution, which simultaneously performs convolution operations in both the spectral and spatial dimensions. This enables the extraction of spectral–spatial features, allowing for the capture of adjacent spatial and channel information in the HSI data. After the feature map is normalized, the ReLU activation function is applied to obtain the output feature map of the 3-D convolution. Subsequently, the output is further processed through a pointwise convolution. The pointwise convolution operates on local areas of the spectral curve, capturing relevant spectral features, or contours of the spectrum. Finally, a 2-D convolution and a pointwise convolution are applied. The 2-D convolutional layer involves sliding a convolution kernel across the spatial dimension, facilitating the extraction of features in local regions of the HSI. The 2-D convolutional layer includes batch normalization and the ReLU activation function. The calculation process of the pointwise convolutional layer is similar to the previous step. A detailed configuration of each convolutional layer in this module is described in Table I.

TABLE I
CONVOLUTIONAL CONFIGURATIONS

Convolutional Layer	Input Channels	Output Channels	Kernel Size	Stride	Padding	BN&ReLU
Conv3D	30	224	(3,3,3)	1	0	✓
Pointwise Conv1	224	224	1	1	0	
Conv2D	224	64	(3,3)	1	0	✓
Pointwise Conv2	64	64	1	1	0	

B. Semantic Tokenizer

The output of the CNN is a set of spectral–spatial features that encode local patterns and characteristics of the input HSI. To better utilize the spectral–spatial features in describing the characteristics of ground objects with a transformer, it is necessary to process the features as tokens. However, the tokens are obtained by flattening the shallow spectral–spatial features from the CNN. The flattening operation limits the ability to maintain context, potentially leading to a loss of semantic coherence. To address the issue, the semantic tokenizer [49] is adopted. Specifically, the flattened spectral–spatial features are represented as $F \in \mathbb{R}^{uv\times z}$, where u denotes the height, v

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represents the width, and z is the number of channels in the spectral–spatial features obtained from the CNN. Then, the semantic tokens $\hat{T}$ can be represent as

$$\hat{T} = (F \cdot W_1) \cdot \text{softmax}(F \cdot W_2)^T \tag{1}$$

where W_1 and $W_2 \in \mathbb{R}^{z \times r}$ are two learnable matrices initialized with a Gaussian distribution. The operation $F \cdot W_1$ maps the spectral–spatial features F to a semantic representation, while the softmax function applied to $F \cdot W_2$ emphasizes important semantics. As a result, r semantic tokens are obtained, denoted as $[T_1, \ldots, T_r]$. These semantic tokens capture the high-level semantic concepts of categories for HSI classification. Finally, by concatenating the trainable class token T^{cls} and adding the positional encodings Pos, the complete semantic token sequence $\boldsymbol{T}$ is obtained

$$\boldsymbol{T} = \left[T^{\text{cls}}, T_1, \ldots, T_r\right] + \text{Pos}. \tag{2}$$

In the transformer architecture, the additional class token plays a role in aggregating information from the entire sequence of semantic patch tokens. However, the single-level class token is limited in its ability to capture interactions with patch tokens for a fixed number of distinct semantics. To address this limitation, we propose an MCTT for HSI classification, which aims to extend the semantic expressive ability of the transformer. The core unit is the cross tokenmixer, which is proposed to enhance the semantic fusion between different stages of tokens, including class tokens and semantic patch tokens. The cross tokenmixer will be detailed in Section II-D.

C. Transformer Encoder

The transformer encoder plays a crucial role in processing the shallow spectral–spatial features by modeling long-range dependencies and capturing global contextual information within the input cubes. The first unit in the transformer encoder is self-attention, which enables the model to attend to different parts of the input features and selectively weigh their importance.

To facilitate self-attention, the input semantic tokens $\boldsymbol{T}$ are transformed into three vectors Q, K, and V through linear transformations using trainable matrices W_Q, W_K, and W_V. At each position in the input sequence, the attention score is computed by taking the dot product between the query vector of that position and the key vectors of all positions. This process generates a set of attention scores that reflect the relevance or importance of each position relative to others.

To assign relative weights to different positions in the input sequence, the attention scores are passed through a softmax function. The values at each position are then multiplied by their corresponding attention scores and summed together, resulting in a weighted sum representation of the values. This process can be described as follows:

$$\text{SA} = (Q, K, V) = \text{softmax}\left(\frac{QK}{\sqrt{d_K}}\right)V \tag{3}$$

where $\sqrt{d_K}$ is a scaling factor used to standardize the dimensions. By incorporating multiple attention heads, the transformer can capture different types of dependencies and patterns within the input sequence. Each attention head can focus on different aspects of the sequence, allowing the model to attend to both local and global information simultaneously. The multiple self-attention heads can be concatenated as follows:

$$\text{MSA} = [\text{SA}_1; \text{SA}_2; \ldots; \text{SA}_m] \tag{4}$$

where m represents the number of multihead attention heads.

In ViT [50], the feed-forward network (FFN) is employed to capture complex nonlinear relationships and refine the representations obtained from the self-attention mechanism. However, for computational efficiency reasons, we utilize the inverted residual FFN (IRFFN) [51] as an alternative. The IRFFN can be represented as follows:

$$\text{IRFFN}(X) = \text{Conv}(\mathcal{F}(\text{Conv}(X))), \tag{5}$$

$$\mathcal{F}(X) = \text{DWConv}(X) + X \tag{6}$$

where Conv represents the convolution operation with the activation layer and batch normalization, and DWConv denotes the depth-wise separable convolution with the activation layer and batch normalization. The utilization of depthwise separable convolution brings benefits in terms of improved parameter efficiency, computational efficiency, and the ability to retain nonlinear transformations. Similar to the classical residual network, the inclusion of a shortcut connection aims to enhance the propagation of gradients across layers.

D. Transformer Encoder With Cross TokenMixer (TECT)

Compared to the conventional transformer encoder, cross tokenmixer is proposed to enhance the interaction between the input and output of the transformer encoder, facilitating the transmission of spectral–spatial features to a hierarchical semantic representation. To distinguish it from the conventional transformer encoder, we refer to this modified version as the TECT block for convenience.

Each TECT block consists of k_i transformer encoders, where i represents the ith TECT block, along with a cross tokenmixer. The input of the cross tokenmixer is divided into two parts: O_i and T_i. The O_i part is derived from the output of the transformer encoder in the ith TECT block. The T_i part functions as the output of the cross tokenmixer in the $i-1$th TECT block. The purpose behind this division is to establish the interactions between the input and output of a series of transformer encoders for hierarchical semantic representation. In addition, O_i and T_i are further divided into class tokens and patch tokens. The class token split from T_i and O_i are denoted as C_i and C_i', while the patch token are denoted as T_i^- and O_i^-. To generate the class token for the final output of the cross tokenmixer, C_i and C_i' undergo linear interpolation, resulting in C_{i+1}. C_{i+1} is the mixed class token for the $i+1$ TECT block, integrating semantic information from different levels of granularity. Moreover, by concatenating T_i^- and O_i^- and subsequently passing them through a fully connected layer, the patch token T_{i+1}^- is generated. Finally, the C_{i+1} and T_{i+1}^- are concatenated to form the output T_{i+1}, which serves as the input to the next TECT block. The output of TECT block T_{i+1}

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can be described as follows:

$$T_{i+1} = \left(w_i C_i + (1 - w_i) C_i'\right) \oplus \mathrm{FC}\left(O_i^- \oplus T_i^-\right) \tag{7}$$

where FC represents the fully connected layer, the symbol $\oplus$ represents concatenation, and w_i represents a learnable parameter utilized in obtaining the interpolated class token through a linear-weighted approach. The length of tokens T_{i+1} in the transformer Encoder increases with i, and the length of T_i is $n = (i+1)^2 + 1$. The expanding number of tokens in TECT enables information transmission across a wider context, facilitating the extraction of coarse-to-fine token representations and promoting feature reuse.

E. Multilevel Class Token Voting Mechanism

Through iterative processing of multiple TECT blocks, local-global semantic information is extracted by modeling the relationship between neighboring sequences. Subsequently, the class tokens from different TECT blocks are employed with the linear layer and softmax function to calculate the probability of the sample belonging to a specific category. Suppose there are M TECT blocks in MCTT, C_M' and $C_i, i = 1, 2 \ldots M-1$ are used for classification

$$c_i = \begin{cases} \mathrm{argmax}(\mathrm{softmax}(W_i \cdot C_i + b_i)), & 1 \le i \le M-1, \\ \mathrm{argmax}(\mathrm{softmax}(W_i \cdot C_i' + b_i)), & i = M \end{cases} \tag{8}$$

where C_i' is the class token from the cross tokenmixer of the ith TECT block, C_i is the class token from the transformer encoder of the ith TECT block, W_i is the weight matrix, b_i is the bias vector and $c_i \in (1, C)$ is the classification category calculated by the class token output from the ith TECT block and C is the total number of classes. Cross-entropy losses are uniformly aggregated across the M branches for training. In the testing phase, the voting count v_j for class j is calculated as follows:

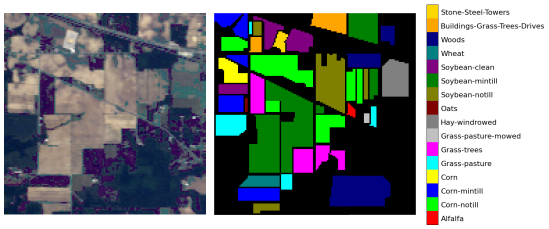
$$v_j = \sum_{i=1}^{M} \delta(c_i, j) \tag{9}$$

where $\delta(c_i, j)$ is an indicator function that equals 1 when c_i is equal to j, and 0 otherwise. The class with the highest votes are regarded as the final classification result

$$\mathrm{v} = \mathrm{argmax}(v_1, v_2, \ldots, v_C). \tag{10}$$

III. Experimental Results and Analysis

This section provides a comprehensive experimental analysis to demonstrate the effectiveness of the proposed MCTT on the HSI classification task. The evaluation of the proposed MCTT was carried out on three benchmark datasets, namely, Indian Pines (IP), WHU_Hi LongKou (WHU-LK), and the University of Pavia (PU). The performance of the proposed method was assessed using three widely used metrics: overall classification accuracy (OA), average classification accuracy (AA), and Kappa coefficient. To ensure robustness, the experiments were repeated ten times with randomly selected training data.



(a) (b)

Fig. 2. IP dataset. (a) False-color map. (b) Ground-truth map.

A. Experimental Datasets

Three publicly available HSI databases, i.e., IP, WHU-LK, and PU are used in the experiments to evaluate the proposed MCTT in the HSI classification.

1) Indian Pines: The IP dataset is a widely utilized HSI classification dataset provided by the Forest Service of the U.S. Department of Agriculture. The dataset comprises 16 distinct land cover categories and a total of 21 025 labeled instances, encompassing common types such as forests, farmland, roads, and urban areas. It is frequently employed to assess the performance of classification algorithms and compare variations between different algorithms. The IP dataset consists of an HSI measuring 145 × 145 pixels and containing 224 spectral bands. Each pixel encompasses 16 attributes, including spectral reflectances measured at different bands. For our experiment, we utilized a rectified dataset that incorporates 200 bands. Fig. 2 displays the false-color composite image, the available ground-truth map, and the scale bar corresponding to the IP dataset.

2) WHU_Hi LongKou: The WHU-LK dataset is a widely utilized HSI classification dataset. It was gathered by the Headwall Nano-Hyperspec imaging sensor over Longkou Town, Hubei, China, which is equipped on a DJ-innovations (DJIs) Matrice 600 Pro UAV platform [52], [53]. The dataset encompasses an HSI with dimensions of 550×441 pixels and consists of 270 spectral bands. The dataset comprises nine distinct land cover classes and a total of 204 542 labeled samples. Fig. 3 displays the false-color composite image, the available ground-truth map, and the scale bar associated with the WHU-LK dataset.

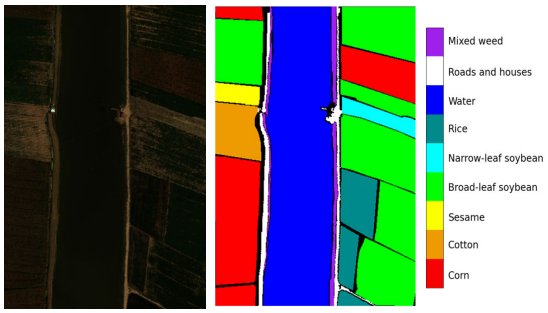
3) University of Pavia: The PU dataset is an additional extensively utilized HSI classification dataset. It was acquired in Pavia, Italy, by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. The dataset comprises an HSI with dimensions of 610 × 340 pixels and encompasses 103 spectral bands. Each pixel is characterized by 103 features, which include spectral reflectances measured at different bands. The dataset encompasses nine distinct land cover classes and a total of 42 776 labeled samples. Fig. 4 displays the false-color composite image, the available ground-truth map, and the scale bar associated with the PU dataset.

Table II presents the names of land-cover categories, as well as the number of training and test samples for IP, WHU_LK, and PU. Each dataset was divided into two sets: a training set and a test set. Specifically, a random 5%, 0.5%, and 2% of the total samples were allocated to the training set for IP,

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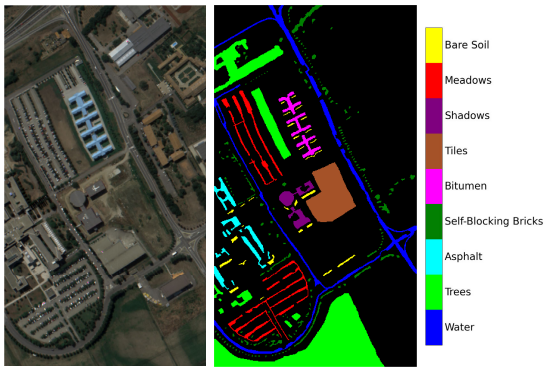
TABLE II
TRAINING AND TEST SAMPLE NUMBERS IN THE IP DATASET, WHU-HI LONGKOU DATASET, AND THE PAVIA UNIVERSITY DATASET

NO.	Indian Pines			WHU_Hi LongKou			University of Pavia		
	Class	Training.	Test.	Class	Training.	Test.	Class	Training.	Test.
1	Alfalfa	2	44	Corn	172	34511	Asphalt	132	6499
2	Corn-notill	71	1357	Cotton	41	8374	Meadows	372	18207
3	Corn-mintill	41	789	Sesame	15	3031	Gravel	41	2057
4	Com	12	225	Broad-leaf soybean	316	63212	Trees	61	3002
5	Grass-pasture	24	459	Narrow-leaf soybean	20	4151	Metal Sheets	26	1319
6	Grass -tree	37	693	Rice	59	11854	Bare soil	100	4926
7	Grass-pasture-mowed	1	27	Water	335	67056	Bitumen	26	1304
8	Hay-windrowed	24	454	Roads and houses	35	7124	Bricks	73	3606
9	Oats	1	19	Mixer weed	26	5229	Shadows	189	9286
10	Soybean-notil	49	923						
11	Soybean-mintill	123	2332						
12	Soybean-clean	30	563						
13	wheat	10	195						
14	woods	63	1202						
15	Buildings-Grass-Tees	19	367						
16	Stone-Steel-Towers	5	88						
	Total	512	9737	Total	1019	204542	Total	1030	59506



(a) (b)

Fig. 3. WHU-LK dataset. (a) False-color map. (b) Ground-truth map.



(a) (b)

Fig. 4. PU dataset. (a) False-color map. (b) Ground-truth map.

WHU_LK, and PU, respectively. This allocation ensured that there was at least one training sample per class, while the remaining samples were used for the test set.

B. Experimental Setting

During the experiments, an initial set of $r = 5$ tokens was utilized as input for the transformer encoder. A batch size of 64 and a neighborhood size of 13×13 were utilized for the input samples. The convergence was accelerated by employing the Adam optimizer with an adaptive learning rate, with an initial learning rate of 0.001. The training process involved 50 epochs for each dataset. The experiments were conducted using a Python 3.9 environment and PyTorch 1.9.1, and the code was executed on an NVIDIA TITAN X graphics card.

C. Influence of Patch Size

The patch size selection in HSI classification plays a critical role in capturing relevant feature information and determining the receptive field of the model. A smaller patch size can lead to information loss, while a larger patch size may include irrelevant information. Thus, it is crucial to choose an appropriate patch size to ensure accurate classification results. Moreover, the chosen patch size also affects the model's focus on local details or broader contextual information. A smaller patch size directs the model's attention toward local details, while a larger patch size enables the capture of broader contextual information. To investigate the impact of different patch sizes, we conducted experiments on three datasets and evaluated the OA for each patch size. The patch sizes selected for evaluation ranged from 9×9 to 17×17. The results, presented in Table III, highlight the influence of different patch sizes on the OA across the three datasets. By keeping other parameters constant, it can be inferred that configuring the patch size as 13×13 yields optimal outcomes for these three datasets. Therefore, selecting a patch size of 13×13 ensures the balance between capturing relevant information and maintaining a suitable receptive field for accurate HSI classification.

D. Influence of Percentage of Training Samples

In this section, we investigated the generalizability of comparison methods for different percentages of labeled training samples. Specifically, for the IP dataset, we randomly selected 2%, 5%, 8%, and 10% labeled samples as training data. For the WHU-LK dataset, the chosen percentages were 0.15%, 0.25%,

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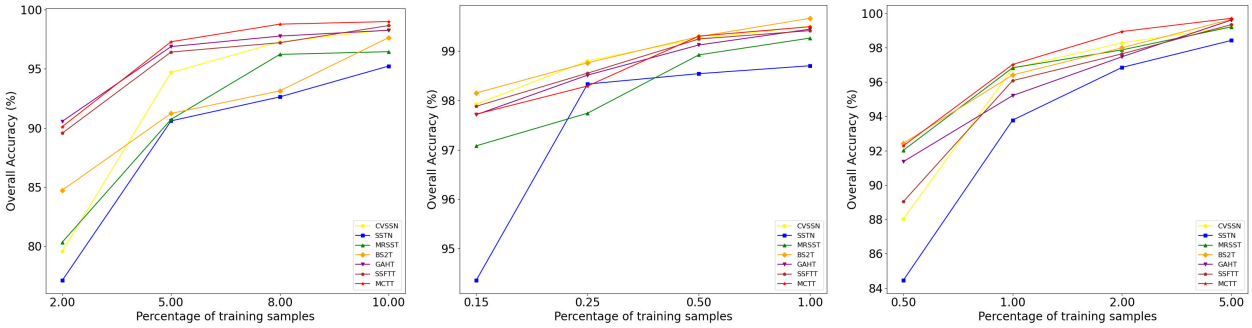


Fig. 5. OA of different methods with different training samples' percentages three datasets. (a) IP. (b) WHU-LK. (c) PU.

TABLE III
IMPACT OF DIFFERENT PATCH SIZES FOR THE OA ON THREE DATASETS

Patch Size	IP	WHU_LK	PU
9*9	94.01	99.08	97.86
11*11	95.94	99.26	98.91
13*13	**97.30**	**99.30**	**98.94**
15*15	96.05	99.14	98.53
17*17	96.52	98.46	98.39

0.5%, and 1%, and for the PU dataset, we opted for 0.5%, 1%, 2%, and 5% labeled samples for training. Fig. 5 illustrates the classification accuracies of several state-of-the-art methods using various training sample specifications. As anticipated, the performance of all methods improves with an increase in the number of training samples. MCTT consistently achieves the highest OA across most cases. In instances where the percentage of training samples is substantial, all methods exhibit elevated OAs. However, exceedingly large sample sizes do not provide discernible reference values in the classification results due to performance saturation. Conversely, when the percentage of training samples is small, notable differences in classification results among various methods become apparent. These findings suggest that MCTT establishes interactions between hierarchical features, effectively capturing more spectral–spatial features than other methods, and demonstrating superior generalization ability.

E. Influence of the Number of TECT Blocks

In this section, we examine the impact of the number of TECT blocks on the classification performance. Fig. 6 showcases the influence of the number of TECT blocks on the OA. Our experiments on the three datasets reveal that the best performance is achieved when the number of TECT blocks is set to 3. Based on our experimental findings, we observed that increasing the number of TECT blocks beyond 3 leads to descent OAs while significantly increasing computational costs. The reason probably lies in the improper granularity of mixed tokens. With the accumulation of TECT blocks, the number of mixed tokens continues to rise, leading to an excessively fine semantic granularity that may hinder the effectiveness of HSI classification.

F. Effect of Mixed Tokens in Cross Tokenmixer

Cross tokenmixer is exploited to establish the interactions between the input and output of transformer encoders in the TECT block. To demonstrate the necessity of the mixed class tokens, the variation of learned w_i across different TECT blocks is investigated. In order to clearly illustrate the variation pattern of w_i, the number of TECT blocks is configured to 5. w_i is the weight to fuse the class tokens from the input C_i and output C_i' of the transformer encoder. The variations of the five parameters (w_1–w_5) were examined, and the corresponding experimental results are presented in Fig. 7. In the initial TECT blocks, as the number of blocks increases, the weight of the class token in the output of the transformer encoders gradually increases, indicating its ability to capture informative spectral–spatial features. When $i = 3$, the weight w_3 of the class token in the transformer encoder output reaches its maximum value, suggesting that the token's granularity aligns perfectly with the requirements of classification, where the suitable semantic representation is captured for the datasets. However, as the number of blocks continues to stack, the value of w_i gradually decreases, indicating that the token's granularity becomes excessively fine, rendering it less apt for classification for the classification task. This phenomenon also further validates that MCTT achieves the best performance at $i = 3$ in Section III-E.

G. Ablation Experiments

In this section, we evaluate the effectiveness of the proposed MCTT by examining various component combinations. The model is divided into three components: spectral–spatial feature extraction, cross tokenmixer, and multilevel class token voting mechanism. To assess the contributions of these components, Table IV presents the classification results of MCTT using different combinations on the three datasets. The baseline is a 3-D CNN combined with a transformer encoder. Spectral–spatial feature extraction refers to the use of spectral–spatial feature extraction block instead of the conventional 3-D CNN in the shallow feature extraction stage. Cross tokenmixer determines whether the input and output of the transformer encoder are mixed in each TECT block. Finally, the multilevel class token voting mechanism determines whether the final classification result is determined

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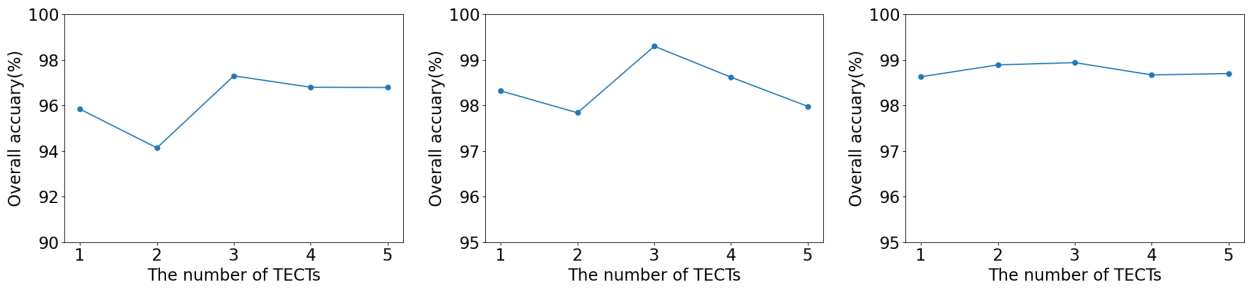


Fig. 6. Impact of the number of TECT blocks for the OA on three datasets. (a) IP. (b) WHU-LK. (c) PU.

TABLE IV
ABLATION ANALYSIS OF THE PROPOSED MODEL ON THE IP, WHU-LK, AND PU DATASETS

case	Components			IP	WHU-LK	PU
	SpectralSpatial Feature Extraction	Cross TokenMixer	Multi-level class token voting mechanism	OA(%)	OA(%)	OA(%)
1				83.17	94.36	85.28
2	✓			93.28	96.83	96.26
3	✓	✓		96.22	99.04	97.70
4	✓	✓	✓	97.30	99.30	98.94

TABLE V
CLASSIFICATION PERFORMANCE WITH DIFFERENT METHODS FOR IP DATASET

class	SVM	1-D CNN	2-D CNN	3-D CNN	CVSSN	SSTN	MRSST	BS2T	GAHT	SSFTT	MCTT
1	48.41±15.9	7.95±3.41	95.56±0.75	26.36±22.11	93.17±5.35	89.16±8.95	57.58±14.88	**98.53±1.62**	23.35±27.20	82.07±18.07	78.18±3.17
2	66.64±4.17	84.56±1.66	77.02±7.52	39.94±16.05	94.60±0.88	88.39±4.18	81.45±2.91	**96.36±2.67**	91.30±3.12	95.14±1.07	95.20±0.88
3	52.37±3.65	45.06±11.47	64.54±21.54	53.11±14.17	95.91±0.93	90.95±3.36	90.75±1.83	95.64±1.93	92.74±2.28	96.20±2.41	**96.83±1.16**
4	42.40±6.80	26.33±14.83	84.59±8.91	29.82±11.15	93.18±4.21	89.29±5.48	78.07±5.68	**99.32±1.26**	60.12±31.36	90.02±6.03	93.16±2.55
5	85.32±3.42	57.63±9.92	11.02±6.23	71.50±11.34	95.75±1.42	92.17±5.71	92.01±2.67	**99.84±0.24**	89.29±3.53	99.39±0.68	99.76±0.21
6	94.13±2.14	98.20±1.23	12.33±12.17	88.73±7.33	97.99±0.79	97.13±1.83	96.86±1.73	**99.42±0.35**	98.92±0.69	98.70±0.58	98.59±0.71
7	52.59±30.76	11.11±11.11	68.58±5.74	11.85±11.80	80.08±9.15	65.98±18.80	62.55±19.08	61.25±7.73	35.62±39.51	**99.59±1.16**	98.89±2.37
8	95.15±3.51	96.56±2.41	62.37±8.46	89.10±10.96	99.28±0.50	99.41±1.01	99.95±0.14	**100.00±0.00**	99.77±0.55	99.36±1.07	99.80±0.29
9	20.00±8.74	10.53±10.53	65.82±15.75	11.58±10.21	80.44±6.53	68.54±16.20	59.06±16.42	80.33±16.86	4.64±10.74	73.10±14.13	**86.32±9.76**
10	64.82±3.49	41.29±25.38	92.32±2.77	38.42±30.83	93.74±1.49	90.25±4.59	90.76±1.93	89.45±3.07	89.59±4.55	96.93±1.41	**97.68±0.78**
11	78.51±2.47	25.50±2.83	86.64±5.19	74.40±8.15	97.09±0.49	95.23±1.08	94.99±1.08	98.45±0.99	92.84±2.46	97.67±2.37	**98.53±0.51**
12	50.85±5.06	24.73±6.48	17.57±5.64	35.28±11.93	90.23±2.52	84.82±6.26	79.88±4.68	**97.79±1.48**	84.59±7.25	88.85±2.08	91.90±1.81
13	93.85±5.07	84.87±10.00	54.62±6.90	93.13±9.08	97.72±1.35	96.98±4.19	99.49±0.80	**100±0.00**	98.41±2.91	100.00±0.00	99.49±0.61
14	93.54±2.21	88.85±5.33	84.88±2.19	90.80±7.70	98.27±0.63	96.94±1.74	98.59±0.88	97.94±0.96	96.99±1.30	99.30±0.37	**99.52±0.39**
15	43.19±24.09	24.11±17.31	48.34±4.18	44.80±22.08	91.18±2.22	87.24±4.31	92.76±2.70	95.11±2.72	84.31±9.79	94.19±3.82	**94.99±2.56**
16	83.86±6.80	84.27±1.12	98.07±0.67	77.98±4.36	**96.33±2.05**	83.15±10.57	95.45±4.29	95.16±3.93	76.09±26.27	78.03±8.26	92.50±4.01
OA(%)	73.77±0.99	56.27±3.82	83.11±1.04	63.54±1.74	95.61±0.35	92.35±1.23	91.44±0.88	96.72±0.4	91.14±1.36	96.50±0.77	**97.30±0.29**
AA(%)	66.60±2.21	50.91±3.33	64.04±2.33	54.80±3.56	93.36±1.24	88.48±1.32	85.64±2.89	94.04±1.15	76.16±4.78	93.03±1.43	**95.08±1.01**
κ*100	69.90±1.15	50.63±3.64	80.14±1.25	58.05±1.67	95.00±0.40	91.28±1.40	90.23±1.01	96.26±0.45	89.88±1.55	96.01±0.88	**96.92±0.33**

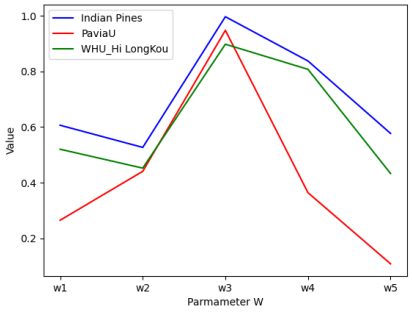


Fig. 7. Variation of parameters w_1–w_5 with the number of TECT blocks for three datasets.

through multilevel class token voting. From the results, it can be observed that the absence of any of the components has a negative impact on performance. In particular, the incorporation of the cross-token mixer results in a notable enhancement, highlighting the effectiveness of incorporating different levels of tokens for HSI classification.

H. Classification Results

To validate the effectiveness of the proposed MCTT, we compared it with several recently proposed classifiers, including SVM [8], 1-D CNN [15], 2-D CNN [16], 3-D CNN [17], central vector oriented self-similarity network (CVSSN) [54], SSTN [46], multirange-spectral–spatial-transform (MRSST) [55], bottleneck SSTN (BS2T) [47], group-aware hierarchical transformer (GAHT) [56], and SSFTT [49]. The methods CVSSN [54], SSTN [46], MRSST [55], BS2T [47], GAHT [56], and SSFTT [49] are considered state-of-the-art methods, with CVSSN relying on CNN and the others incorporating transformer structures. Tables V–VII present the OA, AA, κ, and class accuracy for various methods on the IP, WHU-LK, and PU datasets.

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TABLE VI
CLASSIFICATION PERFORMANCE WITH DIFFERENT METHODS FOR WHU-HI LONGKOU DATASET

class	SVM	1-D CNN	2-D CNN	3-D CNN	CVSSN	SSTN	MRSST	BS2T	GAHT	SSFTT	MCTT
1	98.63±0.26	92.45±6.86	98.27±1.07	98.27±0.60	99.84±0.13	99.50±0.48	99.57±0.31	99.84±0.06	99.85±0.11	**99.93±0.04**	99.92±0.05
2	73.76±2.83	65.73±18.56	75.89±18.43	75.89±10.04	98.50±0.08	96.44±2.12	97.41±1.76	98.77±0.50	97.61±0.96	**99.45±0.30**	99.34±0.52
3	66.98±2.22	66.05±11.60	82.14±10.96	82.14±14.71	99.73±0.21	98.85±0.61	94.60±3.16	**99.74±0.35**	94.54±1.87	95.34±1.35	97.36±1.46
4	96.71±0.38	91.45±5.09	95.47±2.68	95.47±0.71	99.43±0.21	98.74±0.56	99.40±0.17	99.54±0.17	99.46±0.31	99.57±0.14	**99.72±0.10**
5	61.24±2.43	69.17±12.37	76.97±16.96	76.97±6.37	98.16±0.69	97.77±1.26	94.58±2.78	**99.78±0.24**	92.05±2.91	93.37±1.55	95.41±1.94
6	98.75±0.06	97.60±0.84	99.79±0.25	99.79±0.46	99.66±0.22	99.03±0.80	99.03±0.43	**99.88±0.13**	99.78±0.15	99.68±0.23	99.55±0.15
7	99.94±0.02	99.92±0.03	99.94±0.03	99.94±0.04	99.94±0.03	99.87±0.09	99.94±0.04	99.89±0.05	**99.96±0.03**	99.91±0.05	99.87±0.10
8	84.18±1.15	90.44±2.98	82.39±8.70	82.39±3.10	95.21±1.37	90.44±3.16	94.05±1.98	91.30±3.01	95.00±2.56	**95.63±0.57**	95.06±1.56
9	70.52±2.38	71.62±13.09	72.96±14.52	72.96±2.36	91.03±0.89	86.65±4.10	90.52±2.19	**95.52±2.55**	94.06±3.64	92.67±1.27	92.09±2.11
OA(%)	95.01±0.13	92.33±1.58	95.25±1.14	95.26±0.29	99.25±0.11	98.54±0.47	98.92±0.14	99.29±0.08	99.12±0.19	99.24±0.06	**99.30±0.04**
AA(%)	83.41±0.66	82.71±2.92	87.09±3.01	84.27±2.40	97.94±0.24	96.37±1.10	96.57±0.49	**98.25±0.27**	96.92±0.78	97.28±0.39	97.59±0.25
κ*100	93.41±0.18	89.96±2	93.77±1.46	93.74±0.39	99.01±0.14	98.08±0.61	98.58±0.19	99.07±0.11	98.84±0.25	99.01±0.08	**99.07±0.05**

TABLE VII
CLASSIFICATION PERFORMANCE WITH DIFFERENT METHODS FOR PU DATASET

class	SVM	1-D CNN	2-D CNN	3-D CNN	CVSSN	SSTN	MRSST	BS2T	GAHT	SSFTT	MCTT
1	91.05±1.82	91.39±3.95	93.37±1.85	97.71±0.69	97.65±0.39	95.46±1.32	96.74±1.53	96.82±0.95	95.84±0.75	97.00±0.92	**97.90±0.76**
2	94.45±2.65	92.16±3.60	99.31±7.25	98.41±5.95	99.61±0.14	99.34±0.31	99.80±0.12	99.76±0.12	99.91±0.18	99.74±0.76	**99.97±0.03**
3	73.21±2.56	57.80±10.61	91.64±24.86	52.67±25.77	95.51±0.97	92.69±2.91	93.05±2.29	94.41±5.22	89.26±4.41	93.42±1.57	**97.37±0.69**
4	89.05±2.39	89.42±4.18	92.04±3.01	83.42±6.03	**98.36±0.33**	96.66±1.91	95.75±1.23	96.48±0.60	94.25±0.82	95.03±1.67	95.65±1.26
5	99.01±0.22	98.51±1.48	**99.92±0.09**	98.03±1.65	99.72±0.12	99.53±0.50	99.83±0.21	99.90±0.76	99.54±0.02	99.80±0.36	99.73±0.29
6	77.81±5.83	80.54±6.54	80.02±13.81	83.69±18.36	98.85±0.84	98.85±0.84	98.85±0.84	99.12±0.80	99.01±0.91	98.79±0.89	99.25±0.00	**99.95±0.10**
7	77.87±4.92	72.78±13.96	91.72±32.09	67.33±36.05	99.16±0.48	98.46±2.02	97.88±2.09	96.62±0.61	96.80±5.16	87.98±0.34	**99.95±0.11**
8	87.13±1.93	75.85±7.60	89.86±33.31	85.92±20.92	92.52±0.97	85.61±3.46	92.86±3.10	94.27±1.70	94.33±1.54	94.00±2.04	**97.85±0.81**
9	**99.81±0.12**	99.24±0.41	99.14±0.24	98.71±0.69	97.91±0.62	96.86±1.10	95.15±2.42	92.35±1.28	92.74±0.42	94.40±1.34	96.58±1.07
OA(%)	89.65±1.23	87.14±1.62	94.19±0.71	91.21±1.62	98.28±0.10	96.85±0.65	97.87±0.58	97.99±0.36	97.47±0.29	97.64±0.55	**98.94±0.17**
AA(%)	87.71±0.92	84.19±2.27	93.00±1.02	85.10±2.26	97.73±0.11	95.94±0.91	96.69±0.79	96.62±0.42	95.72±0.69	95.62±0.39	**98.33±0.22**
κ*100	86.23±1.60	82.98±2.08	92.22±0.90	88.19±2.08	97.72±0.13	95.82±0.87	97.17±0.77	97.33±0.48	96.64±0.38	96.86±0.73	**98.60±0.23**

Optimal results are highlighted in bold. Among all the methods, the MCTT approach achieves superior classification performance on all three datasets, thereby validating its effectiveness in establishing interactions for hierarchical semantic representation.

In terms of shallow models, SVM [8] performs the worst among all the methods, underscoring the limitations of shallow models in feature extraction for HSI classification compared to deep models. Specifically, 1-D CNN [15] and 2-D CNN [16] methods show poor performance, where only spectral or spatial characteristics are considered. In contrast, the 3-D CNN [17] method, which leverages spectral–spatial features, achieves better results. CVSSN, with its fusion of features using various convolution operations and attention mechanisms to suppress irrelevant features, demonstrates satisfactory performance and enhances the effectiveness of the method. CNNs excel in HSI classification due to their ability to capture spatial and spectral correlations. However, their fixed receptive field size limits their capability to capture long-range spectral–spatial dependencies.

Compared to CNN-based methods, transformer-based methods offer advantages in capturing long-range dependencies and effectively utilizing both spatial and spectral information in HSI classification tasks, thanks to their self-attention mechanism and global context modeling. While transformer-based methods have shown some effectiveness, they have not achieved exceptional performance, as observed in SSTN and MRSST. These methods heavily rely on extracting global features, which can lead to the loss of semantic information during token generation. To address this issue, SSFTT employs semantic tokenization on spectral–spatial features and improves accuracy. However, SSFTT only utilizes high-level class tokens for classification, resulting in a loss of discriminative features and subtle patterns present in the data. In contrast, the proposed MCTT introduces a cross tokenmixer to model the interactions between patch tokens and class tokens across different levels. By integrating the individual contributions of different levels of class tokens, the classification model can capture specific variations and achieve improvements of 2.05%, 0.31%, 2.71% improvement in AA compared to SSFTT on the IP, WHU_LK and PU datasets, respectively.

Furthermore, it is noteworthy that MCTT has achieved favorable accuracy in scenarios with a small number of training samples, such as in classes 1 and 9 of IP. This success is likely attributed to the effectiveness of the semantic tokenizer, which can capture high-level semantic concepts of categories for HSI classification initially, even with limited training samples, before entering the transformer encoder. In addition, the proposed MCTT, employing a voting mechanism based on classification results with varying levels of semantic information, may also ensure accuracy within the constraints of a limited training dataset.

The visualized classification results of IP, WHU-LK, and PU datasets utilizing multiple classification approaches are displayed in Figs. 8–10. Obviously, the MCTT method consistently achieves the most distinct and accurate classification results with clearer boundaries and neater regions, closely resembling the ground-truth labels. In contrast, the other methods exhibit obvious noise in all three datasets, and these issues impact the accuracy of the classification model, affecting its performance.

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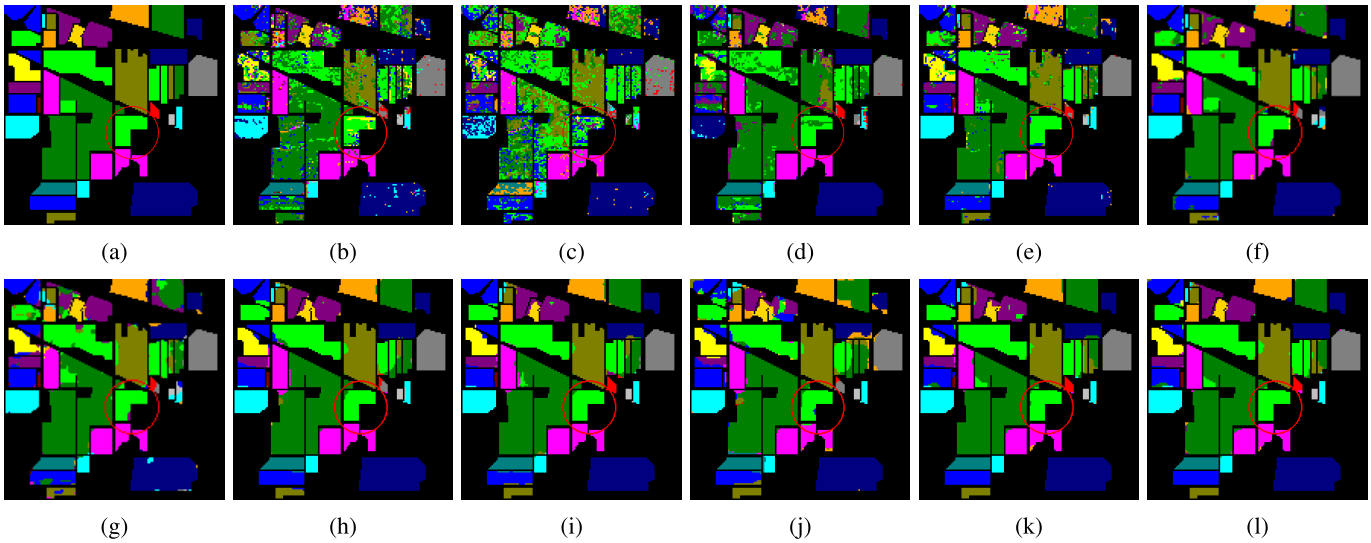


Fig. 8. Classification maps of the IP dataset. (a) Ground-truth map. (b) SVM. (c) 1-D CNN. (d) 2-D CNN. (e) 3-D CNN. (f) CVSSN. (g) SSTN. (h) MRSST. (i) BS2T. (j) GAHT. (k) SSFTT. (l) MCTT.

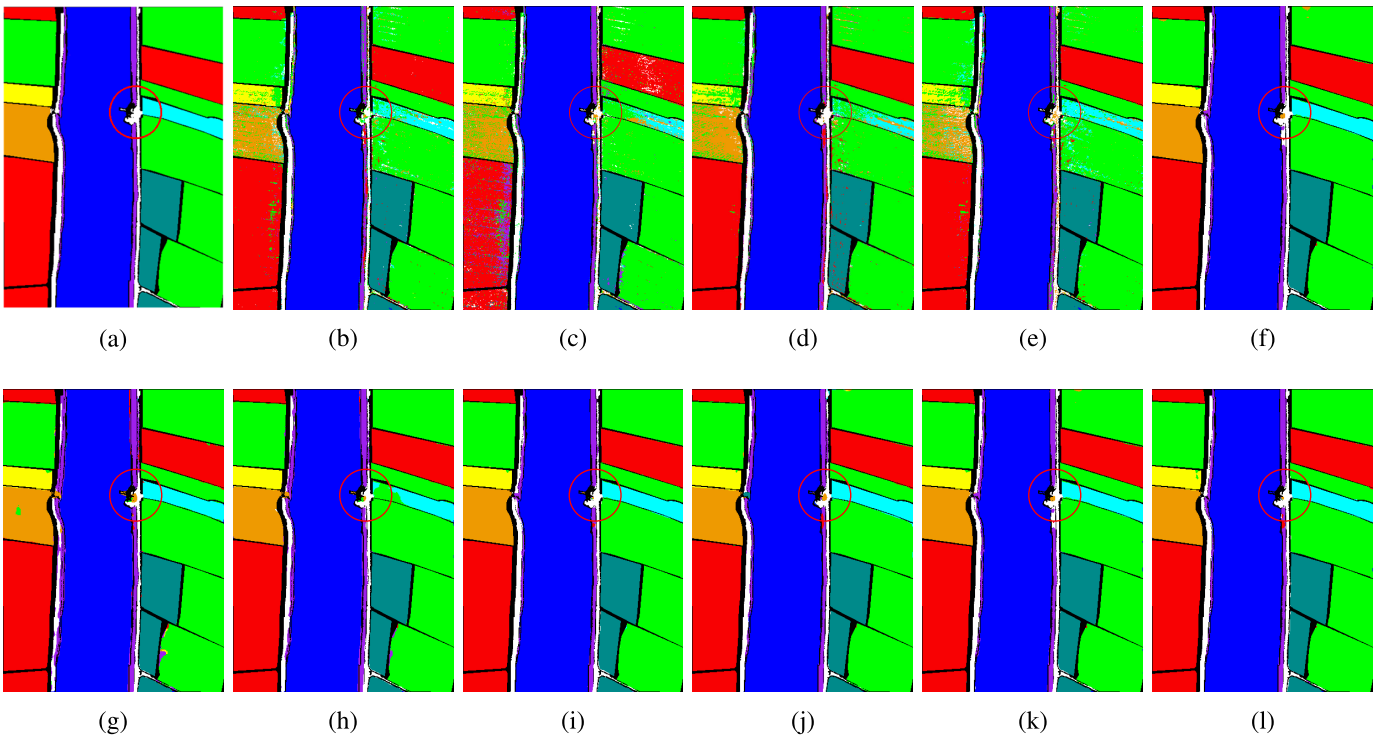


Fig. 9. Classification maps of the WHU-LK dataset. (a) Ground-truth map. (b) SVM. (c) 1-D CNN. (d) 2-D CNN. (e) 3-D CNN. (f) CVSSN. (g) SSTN. (h) MRSST. (i) BS2T. (j) GAHT. (k) SSFTT. (l) MCTT.

I. Complexity Comparisons

In this section, we compare the complexity of the proposed method and other transformer-based methods in terms of running parameter size and floating-point operations (FLOPs). Table VIII shows the results of the model complexity comparison. The proposed MCTT has a higher level of complexity among the compared methods. This is because of the cascaded TECT blocks and the increasing number of tokens, which increase the computation burden. However, the FLOPs and parameter size are within a reasonable range, which enables the proposed MCTT to achieve a balance between classification accuracy and computation efficiency. Nevertheless, model complexity is an important factor to consider, and we will focus on improving the efficiency of the proposed MCTT in our future research.

TABLE VIII
COMPARISON OF MODEL COMPLEXITY OF DIFFERENT TRANSFORMER-BASED METHODS

	SSTN	MRSST	GAHT	BS2T	SSFTT	MCTT
FLOPS(M)	36.81	21.02	886.84	222.41	728.96	862.38
Param(K)	13.17	64.92	202.10	514.11	148.55	437.15

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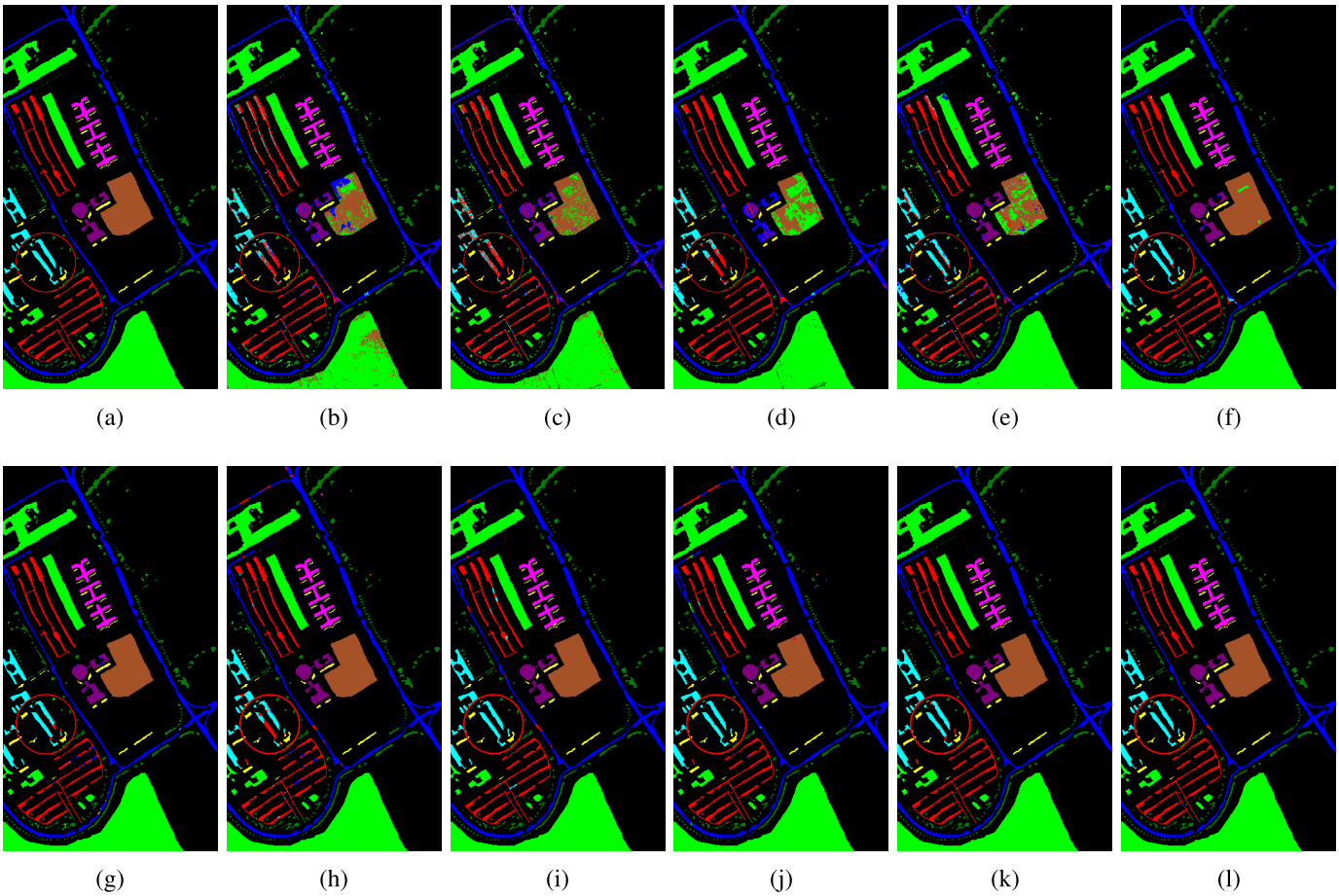


Fig. 10. Classification maps of the PU dataset. (a) Ground-truth map. (b) SVM. (c) 1-D CNN. (d) 2-D CNN. (e) 3-D CNN. (f) CVSSN. (g) SSTN. (h) MRSST. (i) BS2T. (j) GAHT. (k) SSFTT. (l) MCTT.

IV. Conclusion

In this article, an MCTT with cross tokenmixer is proposed for HSI classification. Our proposed MCTT incorporates multilevel tokens to obtain a hierarchal semantic representation of HSI. The convolutional layer is used to capture low-level spectral–spatial features, which are then transformed into semantic tokens. The cross tokenmixer is employed to fuse the input and output tokens of the transformer encoder, creating a hierarchical semantic flow. In addition, we utilize an MCTT architecture to capture high-level semantic features and calculate the probability of pixels belonging to specific categories, thereby improving classification accuracy. Through extensive experimentation on three HSI datasets, we have demonstrated the superiority of our method.

In future research, we plan to investigate a dual-branch network that runs concurrently with both CNN and transformer architectures, leveraging the MCTT to further enhance classification accuracy.

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